MLX91804/5 SW Library

Diagnostic of Acceleration Sensor

Contents

1. Scope

This document explains how the acceleration sensors are diagnosed thanks to the functions Check_IsBadAccelZ and Check_IsBadAccelX in the MLX91804/5 SW library with a revision of 1.10.0 or higher.

2. Acceleration sensor diagnostic

The acceleration sensor diagnostic utilizes internal hardware that applies an electrostatic stimulus to move the sensor's rotor. The resultant displacement is measured as a virtual acceleration. By defining thresholds to the measured values, one can detect the acceleration sensors' mechanical and electrical damages.

2.1. Theory of operation

The voltage, applied to one of the stators with the grounded opposite stator and rotor, creates an electrical field and a corresponding electrostatic force between the stimulated stator and the rotor. Besides, the voltage, applied to the rotor and the opposite stator, creates an electrostatic force in the opposite direction.

Both resultant rotor positions create different capacitance between the rotor and both stators. By measuring both capacitances, it is possible to estimate a total dual rotor displacement from one position to the opposite one. This approach allows cancelling the mechanical offset of the sensor as well as an offset induced by the applied acceleration.

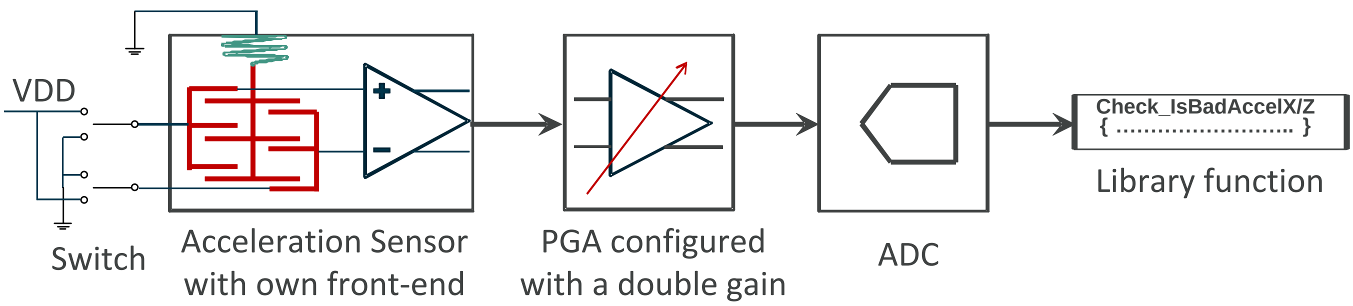

Figure 1: Signal chain at the acceleration sensor diagnostic

To implement the conditions above, the electrostatic stimulus is applied to the sensor via the switch with two opposite polarities (see Figure 1).

To cancel an electrical offset of the signal chain, the responses to the stimulus are also measured with two opposite polarities and with the next subtraction. During the measurements, the Programmable Gain Amplifier (PGA) of the signal chain is configured with a double gain in comparison with the normal acceleration measurements. This way improves twice a signal resolution at a low input acceleration range.

2.2. Voltage dependency

There is a virtual correspondence between the voltage creating displacement and the equivalent mechanical acceleration. **Figure 2** illustrates such a dependency for a typical X-axis acceleration sensor. The Z-axis acceleration sensor behaves similarly.

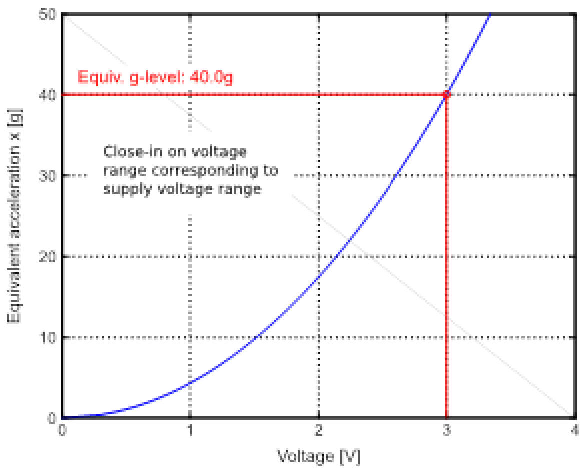

Figure 2: Virtual correspondence between the applied voltage and the equivalent acceleration Done for a typical X-axis acceleration sensor

As is obvious from Figure 1, if VDD = 3 Volt is used as an electrostatic stimulus, the dual rotor displacement corresponds to a virtual acceleration about 40g (with typical parameters of the X-axis acceleration sensor).

2.3. Restrictions

- Because the diagnostic measurements are done with increased PGA gain, it can saturate a signal at high acceleration even within its specified input range. **Thus, the diagnostic cannot be used when a high acceleration is applied to the sensor.**
- Although the implemented diagnostics ignores a permanent acceleration (if it is not large), it is sensitive to an acceleration change that is unavoidable during a driving. **That is why it is strictly recommend using the acceleration sensors' diagnostics at a stationary / parking conditions only.**
- The acceleration sensor has a mechanical stopper limiting the rotor's displacement. It is strongly recommended to avoid the so-called pull-in condition when acceleration and/or the electrostatic force induced by an applied voltage move the rotor up to the mechanical stopper, because potentially it can damage the sensor. The pull-in condition can happen at acceleration that is essentially higher the specified input range. As well, it could happen when the applied electrostatic stimulus (voltage) is close or above 10 Volts, that is not possible within the specified operating range.

3. SW library functions supporting diagnostic

The SW library includes the library functions Check_IsBadAccelZ and Check_IsBadAccelX recommended to verify integrity of Z-axis and X-axis acceleration sensors. The interface of both functions is similar and includes two input parameters defining borders for the virtual acceleration that corresponds with the sensor's response to the electrostatic stimulus induced by VDD voltage applied to the sensor:

bool Check_IsBadAccelZ (int16_t min_threshold, int16_t max_threshold);
bool Check_IsBadAccelX (int16_t min_threshold, int16_t max_threshold);

The input parameters min_threshold and max_threshold have to be fixed by application. Nevertheless, the wide distribution of the acceleration sensitivity complicates defining the reliable thresholds for the response to the applied excitation voltage (VDD). It is especially applicable to the too-sensitive sensors when supplied by VDD from its high operating range.

The implemented characterization allows defining the reliable thresholds for about 99.73% of the produced devices. It corresponds to the acceleration sensitivity range of ±3 sigma of the Gaussian distribution built for 4.2 million devices. The covered devices can be detected and processed according to the table below:

Acceleration sensor	Sensitivity parameter name	nvRAM address	Min	Max	Recommended min_threshold	Recommended max_threshold
X-axis	D_ACCEL_CAL_GAIN_X	0x1064	125	148	400	2500
Z-axis	D_ACCEL_CAL_GAIN_Z	0x1065	92	136	350	3400

Both functions return a Boolean value **"false"** if the acceleration sensor is operable, and **"true"** if it is defective.

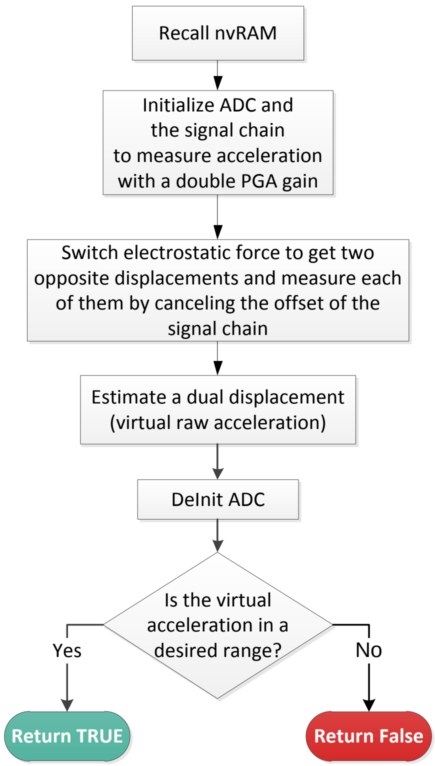

Figure 3: Flow chart of the functions Check_IsBadAccelZ and Check_IsBadAccelX

Note:

- Any measurement error that happened at diagnostic can be detected by the function Adc_IsError.
- Make sure a correct profile is used to build an application. The profile has to match the device type.
- Application has to be in the System mode to call these functions

4. Firmware example description

The firmware example of the acceleration sensor diagnosis is located in the folder **examples/accel_sensor_diagnostics** of the MLX91804/5 SW library.

The example includes a simple **main.c** file that calls the functions **Check_IsBadAccelZ** and **Check_IsBadAccelX** once per second and generates different numbers of pulses on the DEBUG pin GPIO0 depending on the verified sensor's state. The functions are called with the min-max range defined in the table in the section above. The sensors with too high or too low sensitivity are not tested. Note that the final application should not consider the non-tested devices as failed.

The table below specifies the number of GPIO0 pulses generated for different sensors conditions. It is presumed the "accel_sensor_diagnostics" example is built with a proper library profile that matches the used device type:

Available acceleration sensitive axes	State of Z-axis	State of X-axis	Number of generated pulses
Z sensor	Operable or not tested	Not available	1
	Defective	Not available	2
X sensor	Not available	Operable or not tested	3
	Not available	Defective	6
XZ sensor	Operable or not tested	Operable or not tested	1 + 3 = 4
	Operable or not tested	Defective	1 + 6 = 7
	Defective	Operable or not tested	2 + 3 = 5
	Defective	Defective	2 + 6 = 8

The picture below shows the active measurement phases implemented by the "accel_sensor_diagnostics" example for the fully operable and tested XZ acceleration sensor:

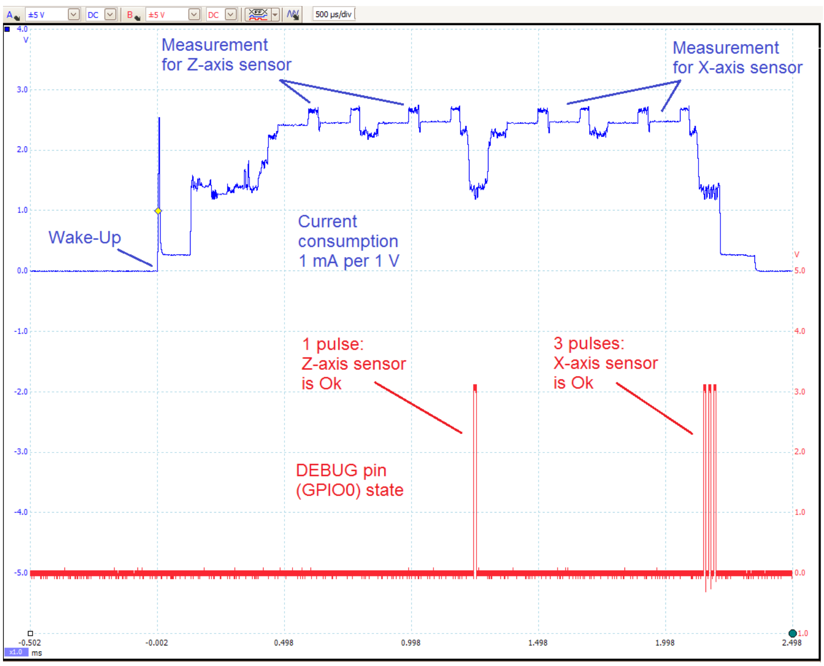

Figure 4: Oscillogram of the acceleration sensors diagnostic done by the functions Check_IsBadAccelZ and Check_IsBadAccelX